SEO Competitor Intelligence Engine

Simple architecture guide — what the engine collects, why it matters, and how the outputs are produced

1. Competitor analysis engine

This part answers: **Why might a competitor rank ahead?** It compares the exact pages ranking for the same keyword, then separates measured facts from interpretation.

Keyword → Google SERP → Confirmed pages → Collect evidence → Validate → Compare → Interpret → Dashboard / PDF

Data source	What it captures	Why it matters
Google SERP / Playwright	Actual ranking positions, ranking URLs and search-result context.	Confirms who really ranks and identifies the exact pages that should be analysed.
Playwright page extraction	Titles, metadata, headings, content, links and structured data.	Shows how each ranking page is presented and what it appears to target.
DataForSEO backlinks	Page-level and domain-level backlink metrics.	Measures authority differences while keeping page evidence separate from domain evidence.
DataForSEO Labs	Related keyword rankings, shared queries and search intent signals.	Shows broader competition and related search coverage.
Lighthouse	LCP, TBT, performance scores and diagnostics.	Finds technical differences worth investigating.
Python engine	Verified calculations, ranking gaps and evidence quality.	Prevents incorrect numbers and unsupported conclusions.
Local Gemma	Evidence-based interpretations and hypotheses.	Explains what differences might mean; it does not calculate the facts.
Dashboard / PDF	Findings, limitations, priorities and actions.	Turns technical evidence into decisions the team can discuss and test.

How a finding becomes an action

Observed difference → Possible explanation → Evidence → Counterevidence → Action → Validation

Important: the engine identifies evidence and plausible explanations. It must not claim that one factor caused Google's ranking decision unless that claim is independently proven.

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Simple architecture guide — how competitor changes are captured and analysed over time

2. Competitor change analysis

This part answers: **What changed on a competitor's page?** It captures verified pages repeatedly, stores each successful version, and compares versions without confusing a failed capture with a real change.

Confirmed URL → Register → Capture → Validate → Store history → Compare → Dashboard → Interpret

Data source	What it captures	Why it matters
Registration	Confirmed URL, keyword and evidence reference.	Ensures that only verified SERP pages are monitored.
Playwright capture	The current rendered webpage state.	Captures what a user and search crawler can see on the page.
SQLite history	Every successful capture with its timestamp.	Makes earlier versions retrievable and auditable.
Snapshot artifacts	An immutable copy of captured page evidence.	Preserves the page state used for the analysis.
Change detector	Differences between successful snapshots.	Finds additions, removals and modifications.
Content hash	A stable fingerprint of page content.	Recognises identical captures and avoids false changes.
Validation	Rendering quality, final URL and extraction status.	Stops failed or partial captures becoming misleading alerts.
Scheduler	Periodic capture runs.	Keeps monitoring continuous instead of relying on manual checks.
Gemma change analysis	Possible SEO or marketing significance.	Explains which changes may deserve human attention.

Fields compared

Page title • Meta description • Canonical URL • Headings • Main content • FAQs • Internal links • External links • Structured data • Promotional messages • Calls to action

Previous	Current	Detected change	Possible response
Play Aviator Online	Play Aviator Online — Get 10 Free Flights	Modified page title	Review title and snippet clarity, then recapture and monitor the target query.